INCH-POUND

MIL-PRF-23145C
19 January 2018
SUPERSEDING
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Amendment 3
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PERFORMANCE SPECIFICATION

INFLATION ASSEMBLIES, LIFE PRESERVER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope: This specification covers the requirements for three types of life preserver inflation assemblies and three different styles of lanyards.

1.2 Classification. The life preserver inflation assemblies and lanyards shall be of the following types and styles, as specified (see 6.2.1):

Type	Lanyard style	Application
I	1	AS6011 cylinders.
II	1, 2 or 3	MIL-C-25369, type II or III cylinders.
III	1 or 3	Dual inflator, AS6011, type I or II cylinders.

2. APPLICABLE DOCUMENTS-

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia, ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or e-mail to dscpg&ispecomments@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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SPECIFICATIONS

FEDERAL

BB-C-101 - Carbon Dioxide (C02) Technical and USP. ,

MILITARY

MIL-C-5040 - Cord, Nylon.
MIL-PRF-6855 - Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes.
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-I-17563 - Impregnates for Cast or Powder Metal Components.
MIL-DTL-19834 - Plates, Identification, Metal Foil, Adhesive Backed.
MIL-C-25369 - Cylinder, Carbon Dioxide, for Inflation of Pneumatic Life Preservers.

STANDARDS FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-130 - Identification Marking of U.S. Military Property.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

62A120H1 - Manifold Stem Assembly and Details, for Life Preservers
975AS121 - Handle, Beaded, Inflation, Life Preserver.

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply

ASTM International Standards:

ASTM A313/A313M	-	Standard Specification for Stainless Steel Spring Wire.
ASTM B85/B85M	-	Standard Specification for Aluminum – Alloy Die Castings.
ASTM B117	-	Standard Specification for Operating Salt Spray (Fog) Apparatus.
ASTM D707	-	Standard Classification System and Basis for Specification for Cellulose Acetate Butyrate Molding and Extrusion Compounds.
ASTM E18	-	Standard Test Methods for Rockwell Hardness of Metallic Materials.

(Copies of these documents are available from http://www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

American Society for Quality (ASQ):

ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes..

(Copies of this document are available online from http://asq.org or from the ASQ Distribution Center, 5131 S. Third Street, Milwaukee, Wisconsin 53207-6028.)

SAE International

AMS-QQ-P-416	-	Plating, Cadmium (Electrodeposited)
AMS-QQ-S-763	-	Steel Bars, Wire, Shapes, and Forgings, Corrosion Resistant.
AS6011	-	Cylinders, Carbon Dioxide Filled, Technical.
AS8660	-	Silicone Compound, NATO Code Number S-736.
AS28775	-	Packing, Preformed, Hydraulic, +275 Deg. F ("O" Ring).

(Copies of this document are available online from http://www.sae.org or the Society for Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-001)

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Uniform Classification Committee, Agent

Uniform Freight Classification Rules

(Copies of this document is available from the Uniform Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1c), the contractor shall furnish sample units for first article inspection and approval (see 4.2 and 6.3).

3.2 Materials and components. Materials and components shall conform to the applicable specifications and drawings as listed or required herein. Unless otherwise specified, the manufacturer shall certify that the materials and components, except for the metallic parts used in the construction of the life preserver inflation assemblies, shall have been manufactured not more than 18 months prior to the date of delivery of the inflation assemblies (see 6.2.2).

3.3 Design and construction. The design and construction of the life preserver inflation assemblies shall be in accordance with figures 1 through 10, as applicable, and applicable drawings.

3.3.1 General design requirements. The life preserver inflation assemblies shall be designed to be compatible with the manifold stem assemblies conforming to Drawing 62A120H1. The inflation assemblies shall consist of a body, with a seat seal gasket (for use with the designated carbon dioxide filled cylinder), two manifold stem gaskets, an internal retractable piercing pin, having an “O” ring and a spring. The assemblies shall also be provided with an outlet having two gas passages, an external actuating lever and a lanyard assembly used for activation of the inflation assembly. “The opening in the body shall contain a chamfered portion and shall be designed to permit the top area of the carbon dioxide filled cylinder to seat firmly against the metal seat of the body. The metal seat shall be smooth, flat and perpendicular to the threads. When not in operation, the lever of the inflation assembly shall rest firmly against the side of the inflation assembly body as shown in figures 1 and 4.

3.3.2 Specific design and construction requirements.

3.3.2.1 Inflation assembly body. The inflation assembly body shall be constructed with a high pressure aluminum die casting conforming to ASTM B85/B85M, alloy A380. The body shall be anodized in accordance with MIL-A-8625, type II class 1 and then impregnated in accordance with MIL-I-17563. The body shall have two gas passages and an outlet for mounting on the manifold stem assemblies conforming to Drawing 62A120H1. One of the gas passages in the outlet shall be aligned with the opening in the side of the manifold stem. Each inflation body shall be threaded with cold rolled internal threads as specified in figures 1 and 4. The threads shall be lightly coated with silicone grease conforming to SAE AS8660.

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3.3.2.2 Gaskets. All gaskets shall be fabricated from synthetic rubber conforming to MIL-PRF-6855, class 2, grade 80 (type optional).

3.3.2.2.1 Manifold stem gasket. The manifold stem shall be provided with two gaskets, one each for the top and bottom of the manifold stem. The top gasket shall be positioned between the body outlet and the cap nut conforming to Drawing 62A120H1-9. The bottom gasket shall be positioned between the body outlet and the manifold stem conforming to Drawings 62A120H1-1 or 62A120H1-2, as applicable. The location and dimensional requirements for the top and bottom gaskets shall be as specified in figure 10,

3.3.2.2.2 Seat seal gasket. The seat seal gasket shall be positioned on the metal seat of the body. The top area of the carbon dioxide filled cylinder shall seat firmly against the upper surface of the seat seal gasket. The seat seal gaskets shall conform to figures 1 or 4, as applicable.

3.3.2.2.3 Set screw. The set screws, for the Type I and III inflation assembles, when applicable, and the Type II inflation assembly, shall be fabricated from corrosion resistant steel conforming to AMS-QQ-S-763, Class 303, Condition A. The threads shall conform to Figures 3 and 5, as applicable, and H-28.

3.3.2.2.4 Type I and III. When specified, the Type I and III inflation assembly set screw shall be a slotted, headless type, with a flat point, conforming to Figure 3. Unless otherwise specified, the Type I and III inflation assemblies shall not contain the set screw.

3.3.2.2.5 Type II. The Type II inflation assembly set screw shall have a slotted head and a flat point, with a nylon insert on the lower portion, conforming to Figure 5. The set screw shall not damage the threads of the carbon dioxide filled cylinder.

3.3.3 Screw threads. The screw threads shall conform to FED-STD-H28 and the requirements specified in figures 1 through 5.

3.3.4 Piercing pins. The piercing pins shall be fabricated with hardened steel, cadmium plated in accordance with AMS-QQ-P-416, type II, class 2. The pins shall have a smooth machine finish, free from tool or chatter marks. They shall be constructed to move in a straight line when activated and shall not waver from side to side. Upon activation, the pins shall retract to their original position. The detail construction of the piercing pins shall be as specified in figures 3 and 7, as applicable.

3.3.4.1 Piercing pin point (for type I and type III). The point of the piercing pin for the type I and type III inflation assemblies shall have four flats. The normal position of the point shall be a minimum of .005 inch below the surface of the metal seat of the body.

3.3.4.2 Piercing pin point (for type II). The point of the piercing pin for the type II inflation assemblies shall have four flats. The normal position of the point shall be a minimum of .100 inch below the surface of the metal seat of the body.

3.3.4.3 Packing, “O” ring. The top portion of the piercing pin shall contain an “O” ring conforming to AS28775, which shall be installed as shown in figures 1 and 4.

3.3.4.4 Spring. The piercing pin shall be supported by a corrosion resistant spring fabricated from wire conforming to ASTM A313/A313M, form I, composition 302, condition B. The diameter of the wire shall be .028 inch. The spring shall be positioned around the piercing pin as shown in figures 1 and 4. The ends of the spring shall be squared and ground.

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3.3.5 Lever. The lever shall be fabricated from hardened steel having a case depth of .010. The steel shall be cadmium plated in accordance with AMS-QQ-P-416, type II, Class 2. The lever shall be securely held in position by a grooved taper pin, as specified in figures 1 and 4. When not in operation, the lever shall rest against the inflation assembly body.

3.3.5.1 Grooved taper pin. The grooved taper pin shall be fabricated from corrosion resistant steel conforming to AMS-QQ-S-763, class 302, condition A and shall conform to figure 6.

3.3.6 Lanyard assemblies.

3.3.6.1 Style 1 and 2 lanyard assemblies. The style 1 and 2 lanyard assemblies shall consist of a continuous length of braided nylon cord conforming to MIL-C-5040, type I (color optional), a plastic knob, and two or three commercial grade brass clips, as required. The knob shall be fabricated from plastic material conforming to ASTM D707, grade H, class optional, black in color. The knob shall be constructed as shown in figure 2. The cord shall be looped through the hole at the end of the lever so that equal lengths extend on each side. The clips shall be .023 +.003 inch in thickness and .375 + .031 inch in length.

3.3.6.1.1 Style 1. The style 1 lanyard assembly shall be as specified in 3.3.6.1. In addition, the two free ends of the cord shall be inserted through the holes in the knob, one end through each hole and shall be tied securely together with a sheet bend knot. Two brass clips, as specified in 3.3.6.1, shall be crimped around the looped cord, one adjacent to the knob and the other adjacent to the lever, to hold the looped cord together. The assembled style 1 lanyard shall conform to figure 2.

3.3.6.1.2 Style 2. The style 2 lanyard shall be as specified in 3.3.6.1. In addition, the style 2 lanyard assembly shall contain a nylon release pin and three brass clips as specified in 3.3.6.1. The release pin shall be fabricated from nylon and shall conform to figure 6. The nylon cord shall be threaded through the hole in the lever. One end of the specified nylon cord, extending from the lever, shall be adjusted to the desired length as specified in figure 6. The longer end of the cord shall then be threaded through one hole in the knob (see figure 6), looped through the second hole in the knob, and subsequently looped through the eye of the release pin as shown in' figure 6. The free ends of the cord shall then be inserted through the holes in the knob, respectively, and securely tied together with a sheet bend knot. The cut ends of the cord shall be seared and no sharp edges shall be formed. The three brass clips shall be crimped around the looped cord, one adjacent to the lever, one adjacent to the release pin, and one adjacent to the knob to hold the looped cord together. The configuration for the style 2 lanyard assembly shall conform in its entirety to figure 6.

3.3.6.2 Style 3 lanyard assemblies. The style 3 lanyard assemblies shall be as specified in Drawing 975AS121.

3.4 Performance characteristics.

3.4.1 Rockwell hardness. The piercing pin and lever, when inspected as specified in 4.5.2, shall have a core Rockwell hardness of C30 to C35.

3.4.2 Weight. The weight of the inflation assemblies, when inspected as specified in 4.5.4, shall not exceed 40 grams for the type I assembly, 55 grams for the type II assembly and 67 grams for the type III assembly.

3.4.3 Leakage.

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3.4.3.1 At 40 psig. The inflation assembly, when inspected as specified in 4.5.5.1, shall show no evidence of leakage. This inspection shall be conducted in sequence with 3.4.3.2. Upon completion of this inspection, the inflation assembly shall be inspected for conformance to leakage at 2 psig, 3.4.3.2.

3.4.3.2 At 2 psig. The inflation assembly, when inspected as specified in 4.5.5.2, shall show no evidence of leakage. In addition, the inflation assembly shall not show any evidence of material or mechanical failure.

3.4.4 Operating force. The inflation assemblies, when inspected as specified in 4.5.6 without the carbon dioxide filled cylinder, shall require a load of not more than 2.0 pounds for effective operation. The assembly shall not be safety wired. The lever shall travel through one normal cycle. The piercing pin shall move in a straight line, shall not waver from side to side, shall advance to its maximum point of travel, and shall retract to its original position.

3.4.5 Piercing pin and lever.

3.4.5.1 Position and travel. The piercing pin, when inspected as specified in 4.5.7, shall perform as specified herein. The normal position of the piercing pin point for the type I and type III assembly shall be as specified in 3.3.4.1. The point of the piercing pin shall travel .100 to .130 inch beyond the surface of the metal seat of the body. The normal position of the piercing pin point for the type II assembly shall be as specified in 3.3.4.2. The point of the piercing pin shall travel .090 to .120 inch beyond the surface of the seat seal gasket. In both cases, the piercing pin point shall be capable of puncturing the carbon dioxide filled cylinder. The lever shall travel through an arc not greater than 115 degrees in line with the longitudinal axis of the body. The piercing pin shall move in a straight line, shall not waver from side to side, and shall retract to its original position.

3.4.5.2 Dimensions of piercing pin and lever. The dimensions of the piercing pins for the types I and III and type II inflation assemblies, when inspected as specified in 4.5.1 and table II, shall conform to figure 3 and figure 7, respectively. The dimensions for the types I and III and type II levers shall conform to figures 1 and 4, respectively.

3.4.6 Operation. The inflation assembly, when inspected as specified in 4.5.8, shall not permit the piercing pin to make any contact with the sealing cap area of the cylinder prior to activation of the lever. The cylinder shall not be punctured nor shall indentation be made. The lever shall not be safety wired. Upon activation of the lever, the piercing pin shall completely operate, and shall advance to its maximum point of travel. The piercing pin shall pierce the center of the puncturable area of the installed carbon dioxide cylinder. The piercing pin shall not stick in the cylinder but shall retract to its original position. The gas in the cylinder shall be released into the atmosphere immediately upon activation and shall not be restricted in any manner. The piercing pin shall not obstruct or retard the flow of carbon dioxide gas in any way, and the gas shall not pass through any part of the inflation assembly into the atmosphere other than through the passages in the outlet.

3.4.7 Hydrostatic pressure. The life preserver inflation assembly, when inspected as specified in 4.5.9, shall not deform or leak. The internal hydrostatic pressure applied shall be 1500 psig.

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3.4.8 Breaking strength of lanyard assembly. The lanyard assembly, when inspected as specified in 4.5.10, shall not break off or fail under a load of less than 70 pounds.

3.4.9 Salt spray. The life preserver inflation assembly, when inspected as specified in 4.5.11, shall show no evidence of corrosion or damage which adversely affects the operation of the assembly. The lever and piercing pin shall not become frozen and shall operate freely. Following the salt spray test, the inflation assembly shall be inspected for conformance to the requirements of 3.4.6.

3.5 Markings. Each inflation assembly shall be marked with legible, durable and sharply defined letters and numerals. The markings shall be molded characters conforming to MIL-STD-130 or inscribed on metal foil identification plates conforming to MIL-DTL-19834, type I, and shall be as follows:

INFLATION ASSEMBLY, LIFE PRESERVER
MIL-PRF-23145C
Contract no.
Manufacturer's name or trademark and part number
Date of manufacture (month and year)

3.6 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with respect to installation and performance.

3.7 Workmanship. The inflation assemblies shall not be misaligned, deformed nor contain any rough surface, nick, burr, sharp edge) crack, embedded foreign matter, detached or loose component, mend, spot, stain, corrosion or scale, pit or dent. The inflation assemblies shall be uniform in quality and free from any mechanical or other irregularities or defects which could adversely affect performance, reliability or durability. The life preserver inflation assemblies shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item meets the requirements of this specification (see 3.1 and 4.2 through 4.2.2).

b. Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots to the requirements set forth in this specification (see 4.3 through 4.3.1.2).

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4.2 First article.

4.2.1 First article samples. Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit twelve inflation assemblies. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of these inflation assemblies from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the acquiring activity (see 6.2.1c). Approval of the first article samples or the waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspection. The first article samples shall be forwarded as specified in the contract or order (see 6.2.1d).

4.2.1.1 First article sample approval. Upon completion of the first article inspection program, recommendations and comments pertinent for use in monitoring production will be forwarded by the Government activity responsible for the inspection program (see 6.2.1d) to the contracting officer. Two of the approved first article inspection sample assemblies will be returned to the manufacturer for use in monitoring production. The other ten will be destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under the contract.

4.2.2 First article inspection. The first article inspection of the life Preserver inflation assemblies shall consist of the examination and tests specified in table I.

4.3 Quality conformance inspection. The sampling and inspection levels shall conform to ASQ Z1.4. The quality conformance inspection shall consist of the following examinations and tests:

Dimensional check of the piercing pins
Visual examination of the inflation assemblies
Dimensional check of the inflation assemblies
Weight
Leakage at 40 psig
Leakage at 2 psig
Piercing pin position and travel and lever operation
Operation with carbon dioxide filled cylinder
Hydrostatic pressure
Packaging

4.3.1 Sampling.

4.3.1.1 Inspection lot.

4.3.1.1.1 Piercing pins. An inspection lot size shall be expressed in units one piercing pin, unfinished (without cadmium plating), for one type of inflation assembly, made essentially under the same conditions and from the same materials. The sample unit shall be one piercing pin, unfinished (without cadmium plating), representing one type of inflation assembly.

4.3.1.1.2 Inflation assemblies. An inspection lot size shall be expressed in units of one inflation assembly of one type made essentially under the same conditions and from the same materials and components. The sample unit shall be one inflation assembly of one type.

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4.3.1.1.3 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing inflation assemblies of one type, fully prepared for delivery from essentially the same materials and components. The sample unit shall be one shipping container, containing inflation assemblies of one type, fully prepared for delivery except that it need not be sealed.

4.3.1.2 Sampling for examinations and tests of the inflation assemblies and packaging. The sample size, acceptance criteria, examinations and tests required for the inflation assemblies and packaging shall be as specified in table II.

4.4 Inspection conditions.

4.4.1 Atmospheric conditions. Unless otherwise specified, all the inspections set forth by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 25° + 10°C (77 + 18°F).

4.4.1.1 Inflation assembly. When the inflation assemblies are inspected at atmospheric pressures or temperatures other than specified in 4.4.1, proper corrections shall be made for the change in instrument readings.

4.4.1.2 Lanyard assemblies. The lanyard assemblies shall be inspected under standard atmospheric conditions of a temperature of 21 + 2°C (70° + 2°F) and a relative humidity of 65 + 2 percent.

4.4.2 Moisture equilibrium. Moisture equilibrium is considered to have been reached when, after free exposure of the material to air in motion controlled at standard atmospheric conditions as defined in 4.4.1.2, and change in weight in successive weighings made at intervals of 1 hour is no greater than .25 percent.

4.4.3 Installation of cylinders.

4.4.3.1 Carbon dioxide filled cylinders. When use of a carbon dioxide filled cylinder is specified in an inspection, the filled carbon dioxide cylinder shall conform to SAE AS6011 or MIL-C-23569 (type II or III) and shall be charged with carbon dioxide conforming to BB-C-101, grade B, type I. The cylinder shall be installed in the inflation assembly, by hand tightening, and shall not make contact with the point of the piercing pin.

4.4.3.2 Empty carbon dioxide cylinders. Unless otherwise specified, the discharged carbon dioxide cylinders conforming to 4.4.3.1, required in conjunction with the leakage and hydrostatic pressure inspections, shall be held firmly in a vise and the inflation assembly shall be tightened to a maximum of 20 inch pounds for the type I and III inflation assemblies and a maximum of 30 inch pounds for the type II assembly.

4.5 Inspection methods.

4.5.1 Dimensional check of the piercing pins. Each of the piercing pins representing one type inflation assembly, selected as a sample unit from the lot, shall be thoroughly checked dimensionally, by use of any suitable measuring device, to determine conformance to this specification. The list of defects, table III, shall be used to enumerate the defects found.

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4.5.2 Rockwell hardness. The Rockwell hardness of the piercing pin and lever, as applicable, shall be determined in accordance with ASTM E18, C scale. At least three hardness determinations shall be made on the piercing pin and lever, as applicable. For the piercing pin, the hardness shall be determined at the head, center and near the point. For the lever, the hardness shall be determined on the flat side that is near the portion which makes contact with the head of the piercing pin. The hardness shall be determined at a minimum distance of .250 inch apart. The three hardness determinations for each piercing pin and lever, as applicable, shall be averaged.

4.5.3 Visual examinations.

4.5.3.1 Inflation assemblies. Every inflation assembly shall be examined visually for major defects to determine conformance to this specification. Each inflation assembly, selected as a sample unit from the lot, shall be examined visually for minor defects and thoroughly checked dimensionally to determine conformance to this specification. The classification and list of defects, tables IV and V shall be used to classify and enumerate the defects found.

4.5.3.2 Packaging. Each of the fully prepared shipping containers, containing inflation assemblies, selected as a sample unit from the lot, shall be examined visually to determine that the packaging, packing and marking conform to this specification. The list of defects, table VI, shall be used to enumerate the defects found.

4.5.4 Weight. The weight of the inflation assembly shall be determined on any scale or balance capable of weighing to the nearest .1 of a gram.

4.5.5 Leakage.

4.5.5.1 40 psig. An applicable discharged carbon dioxide cylinder shall be installed in the inflation assembly as specified in 4.4.3.2. The lever shall be in the actuated position during this inspection. The actuated position is considered to be the position of' the lever after the piercing pin has punctured the carbon dioxide filled cylinder as specified in 4.5.8. The inflation assembly shall be immersed in water and air pressure of 40 psig, free of oil and condensed water vapor, shall be applied at the outlet of the inflation assembly for a minimum of 10 seconds. While the pressure is applied, the inflation assembly shall be observed for leakage. Upon completion of this inspection, the pressure shall be reduced to zero and the inflation assembly shall be subjected to the 2 psig leakage inspection, 4.5.5.2.

4.5.5.2 2 psig. An applicable discharged carbon dioxide cylinder shall be inserted in the inflation assembly as specified in 4.4.3.2. The lever shall be in the unoperated position. The inflation assembly shall be immersed in water and air pressure of 2 psig, free of oil and condensed water vapor, shall be applied at the outlet of the inflation assembly for a minimum of 10 seconds. While the pressure is applied, the lever shall be slowly moved through at least one complete normal operation and the inflation assembly shall be observed for leakage. Upon completion of this inspection, the inflation assemblies shall be dried without the application of heat.

4.5.6 Operating force. For this inspection, no carbon dioxide cylinder is required. The inflation assembly shall be held in a vertical position, with the opening at the top, or in a horizontal position, with the lever on the bottom. The lever shall be in the normal position and a maximum load of 2 pounds shall be attached to the free end of the lever. The load shall operate the lever through one normal cycle and the travel, movement and motion of the piercing pin shall be observed for conformance to 3.4.5.1.

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4.5.7 Piercing pin position and travel and lever operation. An accurately calibrated suitable measuring device shall be used to determine the requirements of 3.4.6. If desired, the seat seal gasket may be removed for this inspection. During this inspection, the movement and motion of the piercing pin shall also be observed for conformance to 3.4.5.1.

4.5.8 Operation. For this inspection, a carbon dioxide filled cylinder conforming to SAE AS6011 or MIL-C-25369, type II or III, as applicable, shall be fully installed in the applicable inflation assembly with hand tightening. The inflation assembly shall be examined to assure that the top of the cylinder has not made contact with the point of the piercing pin by listening for the sound of escaping gas. The carbon dioxide filled cylinder shall then be removed and the top of the cylinder shall be examined. The cylinder shall not have been punctured and there shall not be any evidence of indentation by the piercing pin. Upon completion of the examination, the carbon dioxide cylinder shall be fully reinstalled in the inflation assembly with hand tightening. The inflation assembly shall be held in the vertical position, with the cylinder at the top, or in a horizontal position, with the lever on the bottom. The lever shall be in its normal position and a maximum load of 15 pounds shall be attached to the free end of the lever. The load shall operate the lever through one normal cycle. The piercing pin shall advance to its maximum point of travel and shall puncture the carbon dioxide filled cylinder in the center of the puncturable area. The piercing pin shall not stick in- the carbon dioxide filled cylinder but shall retract to its original position. The gas shall be immediately released and the piercing pin shall not obstruct or retard the flow of the gas. The gas shall not pass through any part of the inflation assembly into the atmosphere other than through the gas passages in the outlet. The empty cylinder shall be removed and the top shall be examined for the location of the puncture.

4.5.9 Hydrostatic pressure. An applicable discharged carbon dioxide cylinder shall be installed in the inflation assembly as specified in 4.4.3.2. The lever shall be in the actuated position during this inspection. The actuated position is considered to be the position of the lever after the piercing pin has punctured the carbon dioxide filled cylinder as specified in 4.5.6. A hydrostatic pressure of 1500 psig shall be applied to the outlet of the inflation assembly for a minimum of 5 minutes. While the pressure is applied, the inflation assembly shall be observed for leakage and deformation.

4.5.10 Breaking strength of the lanyard assembly. The breaking strength of the lanyard assembly shall be determined by securing the lever to the upper jaw of a suitable inspection apparatus and the knob, in a suitable jig, shall be attached to the bottom jaw. The portion of the lever containing the cord shall be between the upper and lower jaws. The speed of the inspection apparatus under no load shall be 12.000 + .500 inches per minute. The jaws shall be separated until failure occurs. When applicable, the cord, looped through the eye of the release pin of the type II inflation assembly, shall hang free and shall not be attached to any jaw of the inspection apparatus.

4.5.11 Salt spray. The inflation assembly shall be exposed to salt spray for 100 hours in accordance with ASTM B117. The inflation assembly shall be exposed in a vertical position with the opening at the top and in a manner that the conditioned air circulates completely around the inflation assembly. Upon completion of the salt spray, the inflation assembly shall be examined for conformance to 3.4.9. Upon completion of this examination, the inflation assembly shall be subjected to the operation with the carbon dioxide filled cylinder inspection, 4.5.8.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The inflation assemblies covered by this specification are intended for use as a component of the mechanical inflation systems used for inflating life preservers.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type of assembly, style of lanyard, type of cylinder and quantity desired. (see 1.2).
c. Whether first article inspection is waived. (see 4.2.1).
d. Name and address of the first article inspection laboratory (see 4.2.1) including the Government activity responsible for conducting the inspection program (see 4.2.1.1).
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
f. Whether any special markings are required (see 5.2).

6.2.2 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
3.2	Certificate of compliance for the age of materials and components	DI-E-2121 - Certificate of compliance
4.2.1.1	First article inspection reports	DI-T-5329 – Inspection test reports

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer).

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6.3 First article. When first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations. The first article should be a first production item. The first article should consist of twelve units. The contracting officer should include specific instructions in all acquisition instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Subject term (key word) listing.

External actuating lever
Lanyard assembly
Piercing pins
Two gas passages

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

TABLE I. First article examinations and tests.

Inspection	Paragraph	
	Requirement	Method
Visual examination	3.3 through 3.3.6.2, as applicable, applicable drawings and figures	4.5.3.1
Rockwell hardness	3.4.1	4.5.2
Weight	3.4.2	4.5.4
Leakage at 40 psig	3.4.3.1	4.5.5.1
Leakage at 2 psig	3.4.3.2	4.5.5.2
Operating force	3.4.4	4.5.6
Piercing pin position and travel	3.4.5.1	4.5.7
Operation	3.4.6	4.5.8
Hydrostatic pressure	3.4.7	4.5.9
Breaking strength of lanyard	3.4.8	4.5.10
Salt spray	3.4.9	4.5.11
Dimensional check	3.3 through 3.3.6.2, as applicable, 3.4.5.2 and applicable figures and	4.5.1 and 4.5.3.1

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TABLE II. Sample size, acceptance criteria, examinations and tests of the inflation assemblies and preparation for delivery

Inspection	Paragraph		Sample size	Acceptance criteria
	Requirement	Method		
Dimensional check of the piercing pins	3.4.5.2 and figures 3 and 7, as applicable	4.5.1	Inspection level S-3	An acceptable quality level of 2.5 defects per 100 units.
Visual examination of the inflation assemblies	Section 3 and applicable figures.	4.5.3.1	Every inflation assembly for major defects and inspection level II for minor defects	Reject all units with any major defect and an acceptable quality level of 4.0 defects per hundred units for minor defects.
Dimensional check of inflation assemblies	Section 3 and applicable figures	4.5.3.1	Inspection level S-3	An acceptable quality level of 2.5 defects per 100 units
Weight	3.4.2	4.5.4	Inspection level S-1	An acceptable quality level of 2.5 defects per 100 units
Leakage at 40 psig	3.4.3.1	4.5.4.5.1	Every inflation assembly	Reject all defective units
Leakage at 2 psig	3.4.3.2	4.5.5.2	Inspection level S-2	An acceptable quality level of 1.0 defects per 100 units
Piercing pin position and travel and lever operation	3.4.5.1	4.5.7	Every inflation assembly	Reject all defective units
Operation with carbon dioxide filled cylinder	3.4.6	4.5.8	Inspection level S-2	An acceptable quality level of 1.0 defects per 100 units
Hydrostatic pressure	3.4.7	4.5.9	Inspection level S-1	An acceptable quality level of 2.5 defects per 100 units
Packaging	Section 5	4.5.3.2	Inspection level S-2	An acceptable quality level of 2.5 defects per 100 units

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TABLE III. List of defects for the finished dimensions of the piercing pins.

Examine	Defect
Check the various dimensions of the piercing pin as specified in figures 3 and 7, as applicable, and the “O” ring.	Any measurement deviating from 3.3.4 through 3.3.5 and figures 3 and 7, as applicable, and applicable tolerances shall be enumerated as dimensional defect

TABLE IV. Classification of defects for the visual examination of the inflation assemblies.

Major	
101.	Any surface rough, misaligned or contains any nick, unspecified sharp edge, crack, burr, pit, dent, sliver, scale, embedded foreign matter, loose particle or other imperfection.
102.	Any finish missing or any component improperly finished.
103.	Any component missing, malformed, corroded, fractured, chipped, bent, distorted or damaged.
104.	Any component loose, detached, not assembled as specified or otherwise not securely retained.
105.	Any end of the grooved taper pin protruding from the body.
106.	Any functional part that works with difficulty.
107.	Any gas passage missing.
108.	Any threads missing, stripped, torn, broken or otherwise defective.
109.	Any component not as specified or any defect of a component or defect of assembly not herein classified which seriously affects the serviceability or appearance.
110.	Any component, component part or required operation omitted or any operation not herein classified improperly performed thereby seriously affecting serviceability or appearance.
Minor	
201.	Any markings, missing, incorrect, illegible, incomplete, not of a permanent nature or improperly located.
202.	When applicable, contract number and date of manufacture not coated (see 3.5).
203.	Surface unclean; or foreign matter appearing on surface.
204.	Any component not as specified or any defect of a component not herein classified which does not seriously affect the serviceability or appearance.
205.	Any component, component part or required operation omitted or any operation not herein classified improperly which does not affect serviceability or performance.

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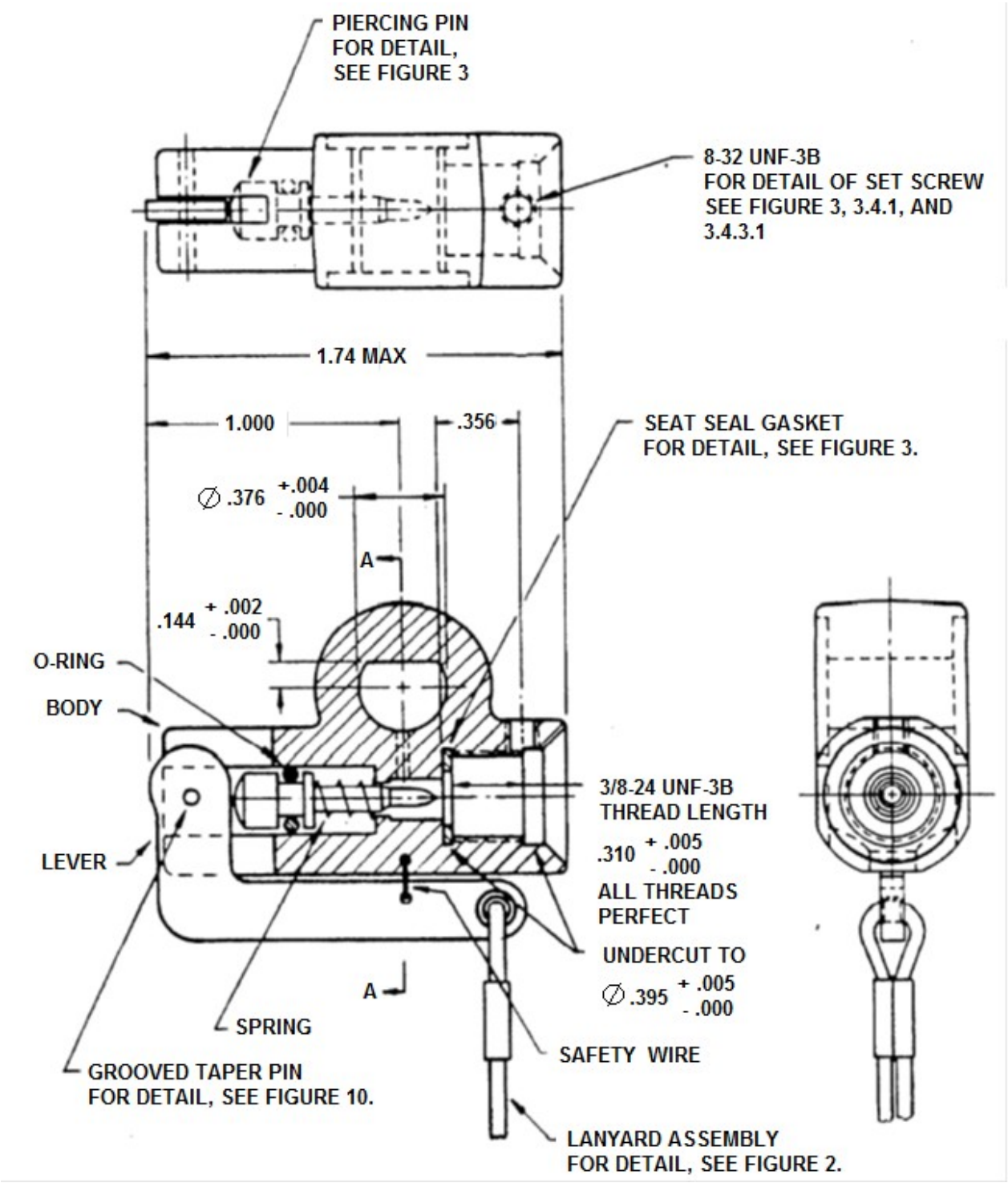
TABLE V. List of defects for the finished dimensions of the inflation assemblies.

Examine	Defect
Measure the assembly outline, outlet, body, thickness of the lever, set screw, when applicable, lanyard assembly, knob, threads and top and bottom gaskets.	Any measurement deviating from the dimensions specified in figures 2 through 8 and allowable tolerance, as applicable shall be enumerated as a dimensional defect.

TABLE VI. List of defects for preparation for delivery.

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence or method of application; markings not the same on the interior and exterior containers; information on the paper and label in the polyethylene bag not legible through the film or no label in the film bag.
Packaging and packing materials	Any non-conforming component; any missing damaged or otherwise defective.
Workmanship	Inadequate application of the components; incomplete closure of the polyethylene bag, unit or intermediate package, container flaps or loose strapping; bulging or distortion of the containers. Any portion of the lanyard assembly caught in any of the heat sealed seams.
Exterior and interior weight or content	Gross or net weight exceeds the requirements; more than one type of inflation assembly packed in the same container.

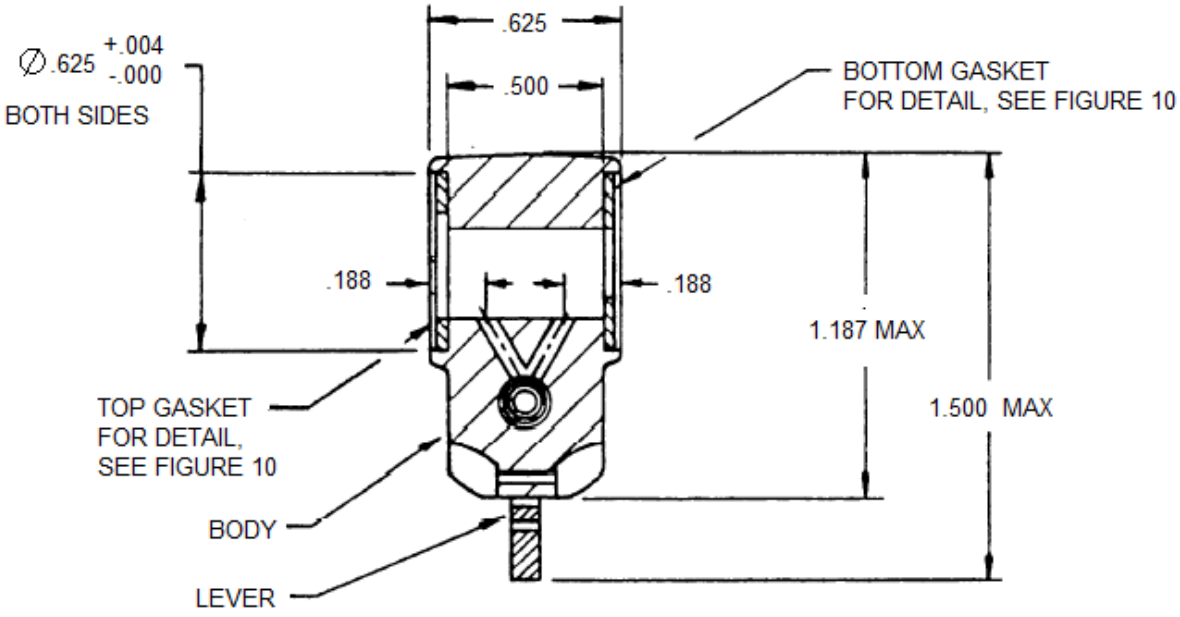
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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED.
DECIMALS ± .005
TOLERANCES: FRACTIONS ±1/64,
FOR SECTION A-A, SEE FIGURE 2

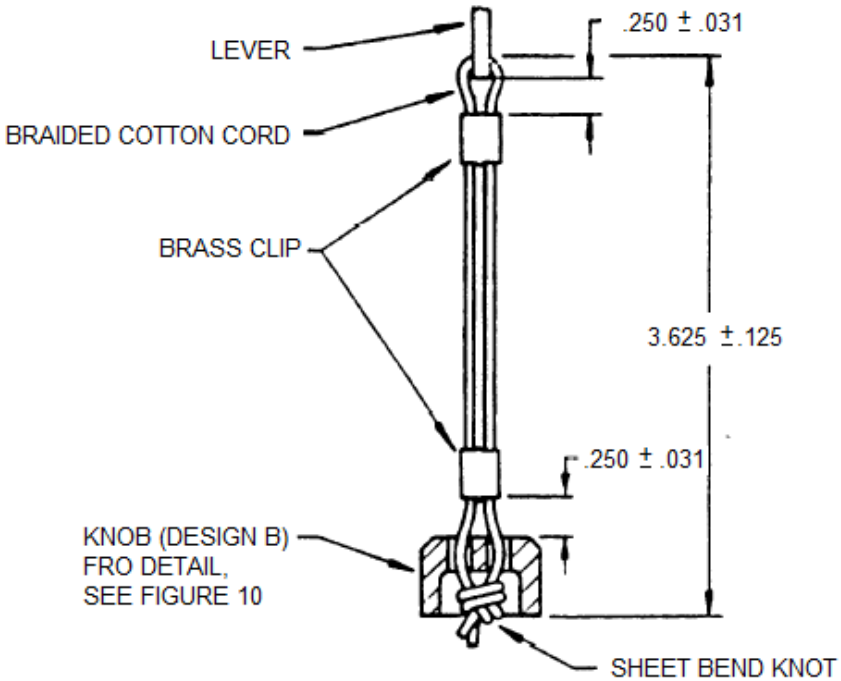
FIGURE 1. TYPE I INFLATION ASSEMBLY

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SAFETY WIRE OMITTED

SECTION A - A



DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE AS SPECIFIED IN FIGURE 1

LANYARD ASSEMBLY

FIGURE 2. SECTION A-A AND LANYARD ASSEMBLY FOR TYPE I INFLATION ASSEMBLY

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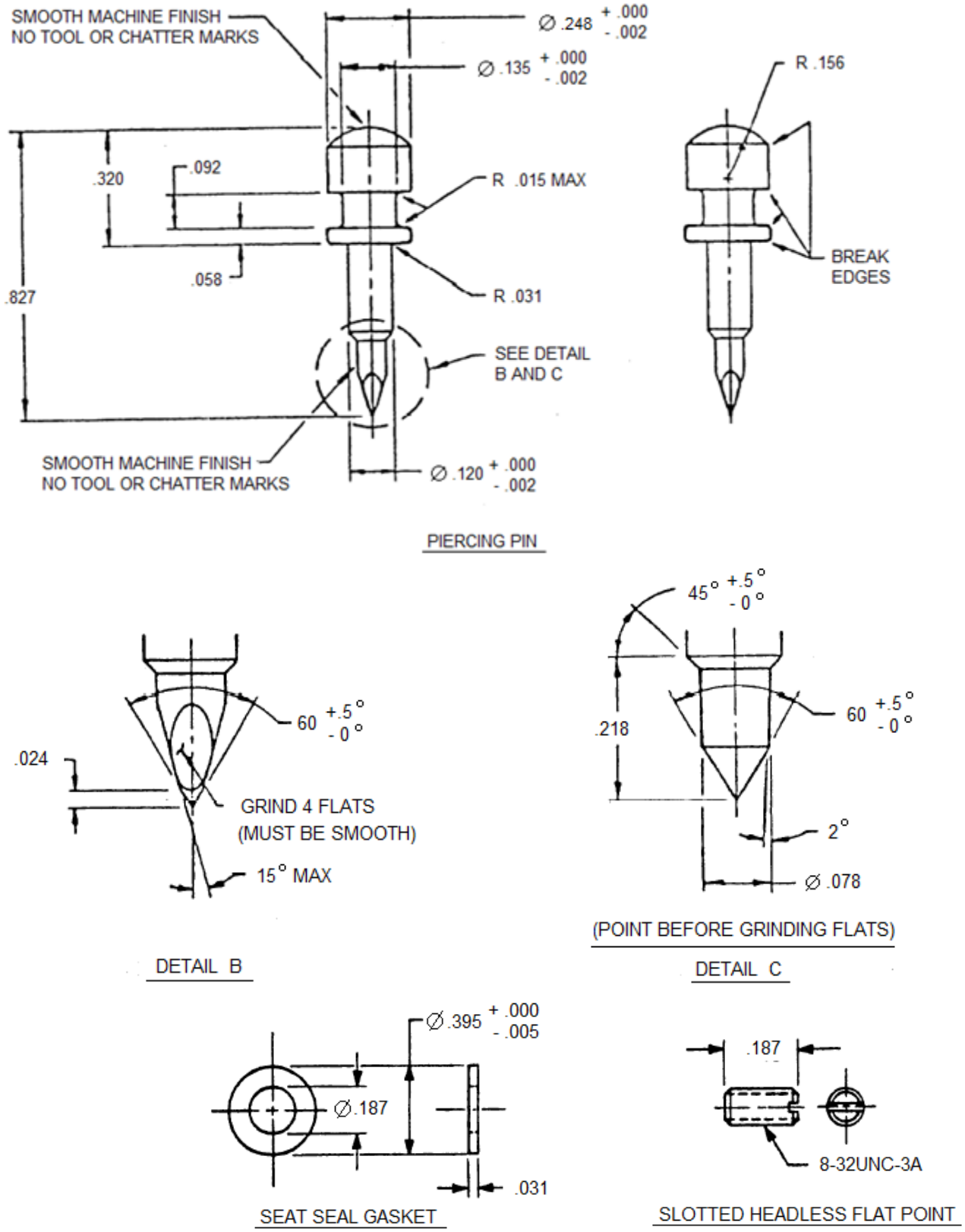


FIGURE 3. PIERCING PIN, SEAT SEAL GASKET, AND SET SCREW RETAILSFOR TYPE I AND TYPE III INFLATION ASSEMBLIES.

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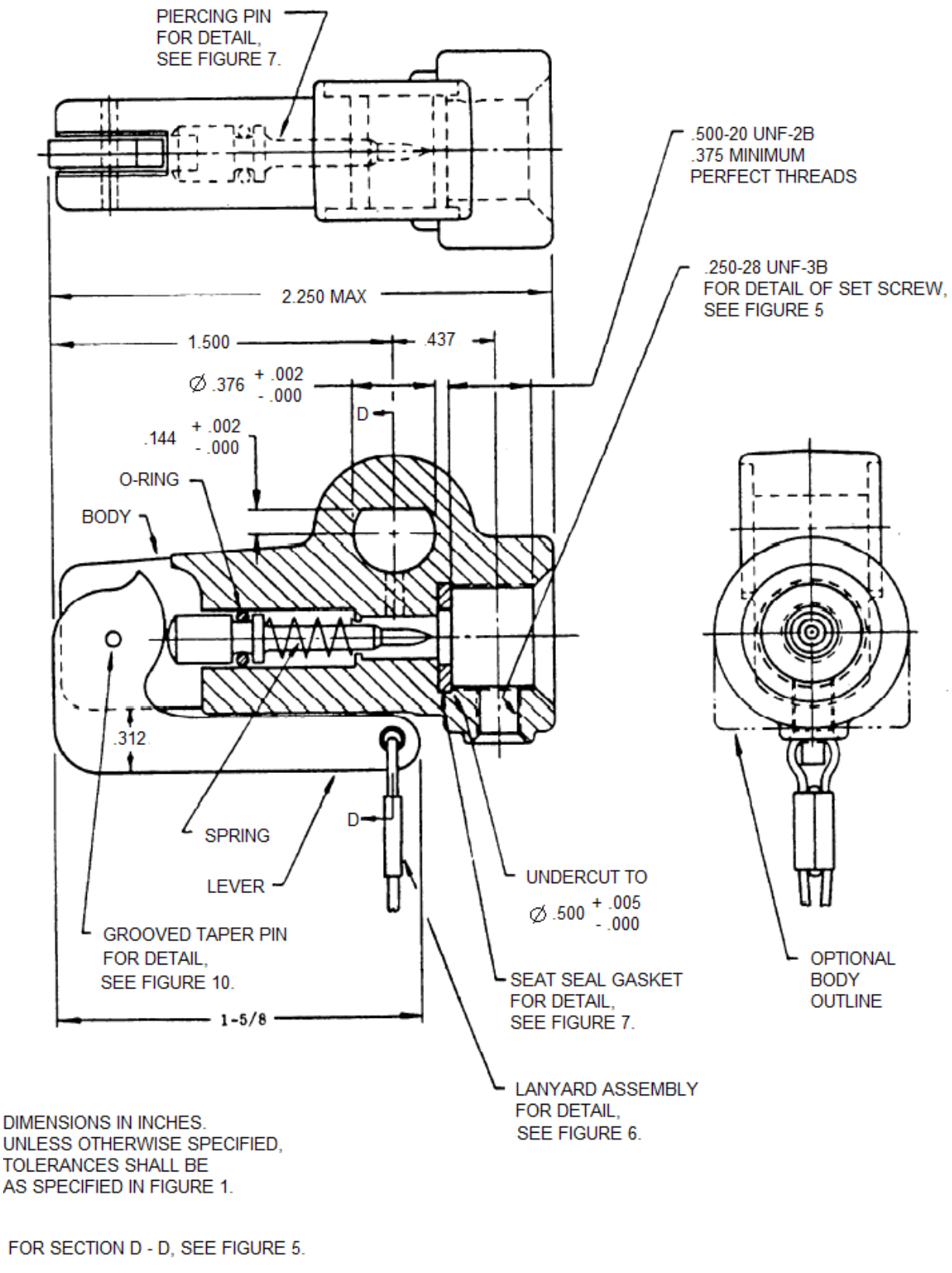
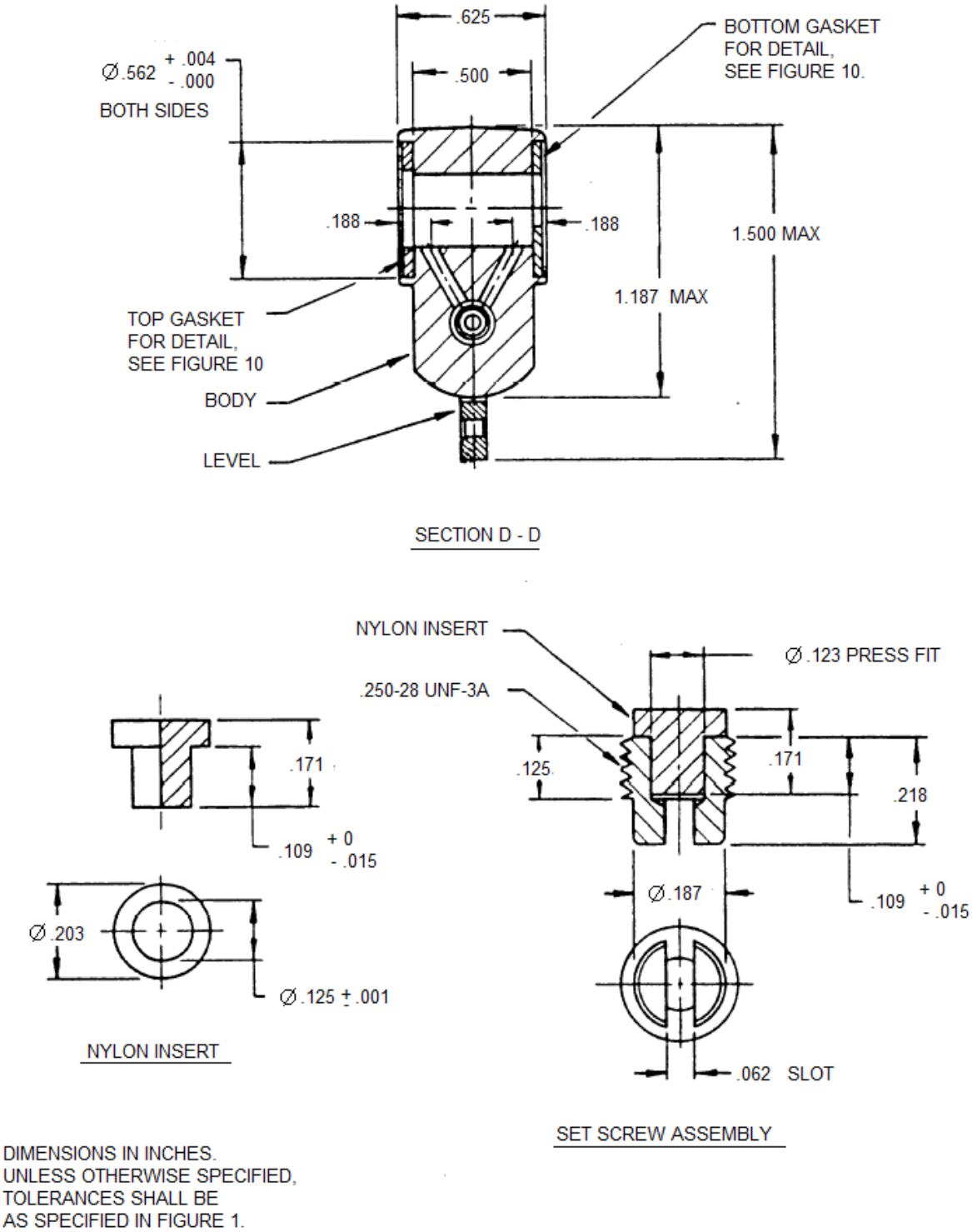


FIGURE 4. TYPE II INFLATION ASSEMBLY

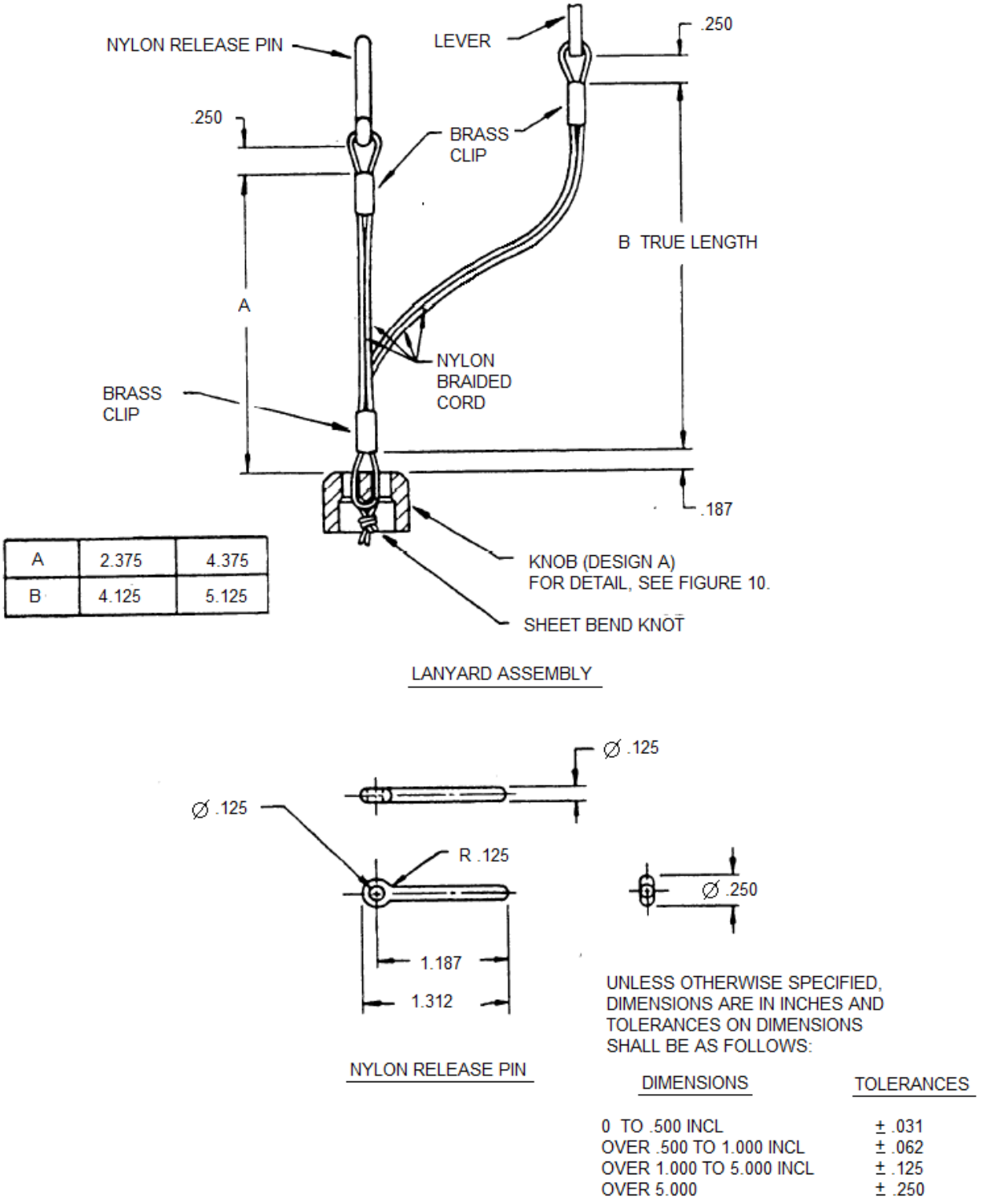
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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES SHALL BE
AS SPECIFIED IN FIGURE 1.

FIGURE 5. SECTION D - D, NYLON INSERT, AND SET SCREW ASSEMBLY DETAILS FOR TYPE II INFLATION ASSEMBLY

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A	2.375	4.375
B	4.125	5.125

FIGURE 6. LANYARD ASSEMBLY AND NYLON RELEASE PIN DETAILS FOR TYPE II INFLATION ASSEMBLY

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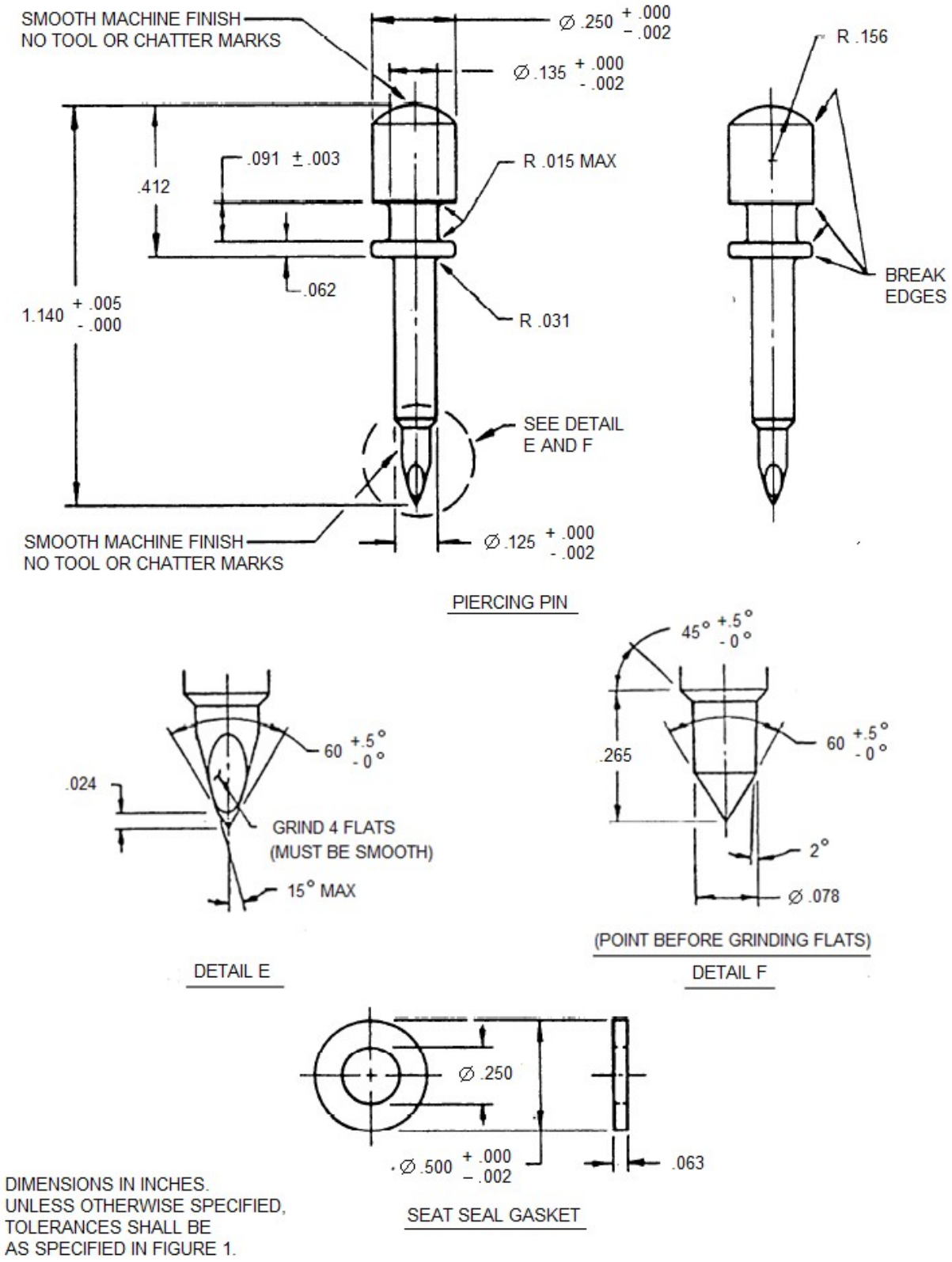
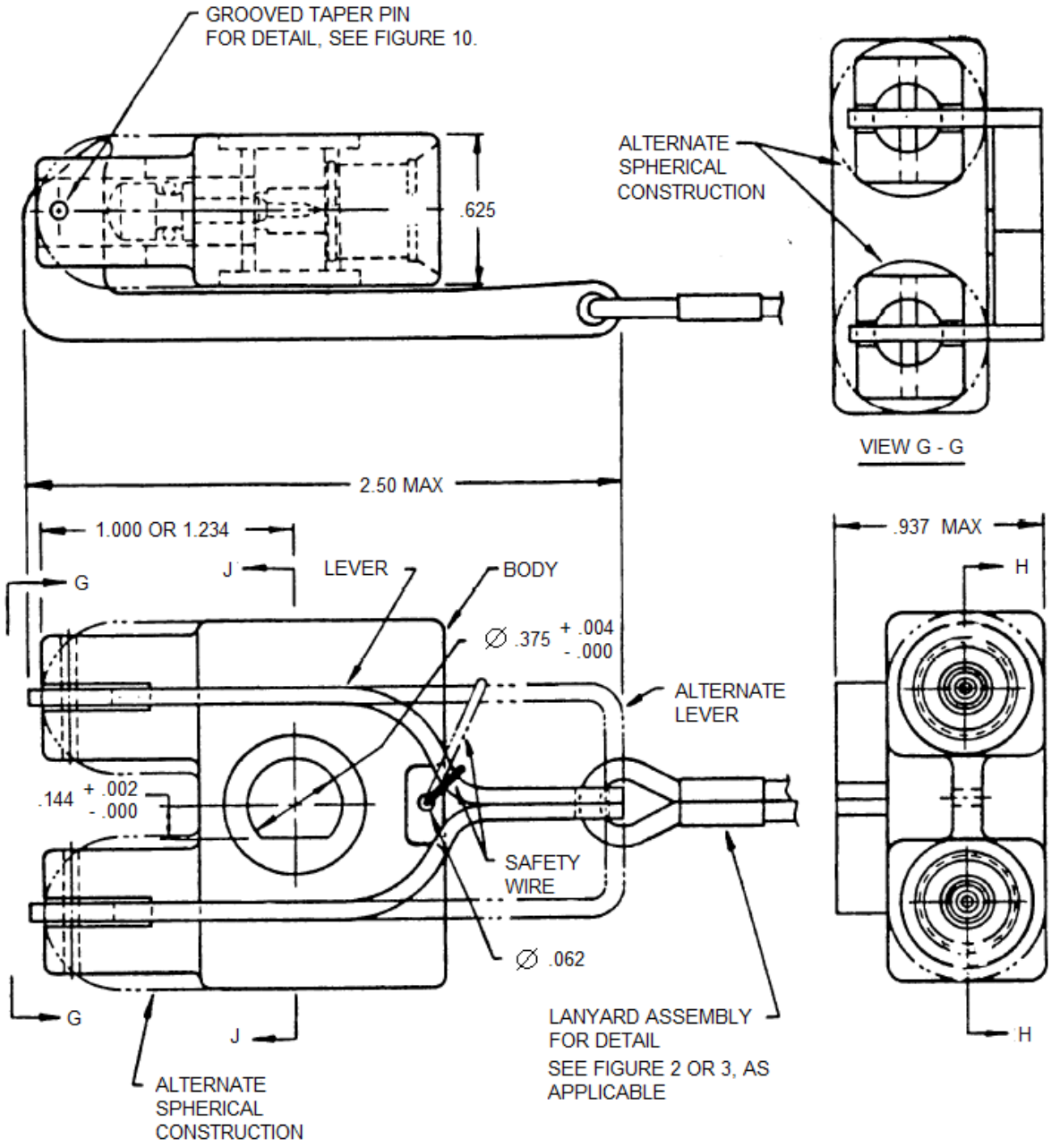


FIGURE 7. PIERCING PIN AND SEAT SEAL GASKET DETAILS FOR TYPE II INFLATION ASSEMBLY

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES SHALL BE
AS SPECIFIED IN FIGURE 1.

FOR SECTIONS H - H,AND J - J, SEE FIGURE 9.

FIGURE 8. TYPE III INFLATION ASSEMBLY

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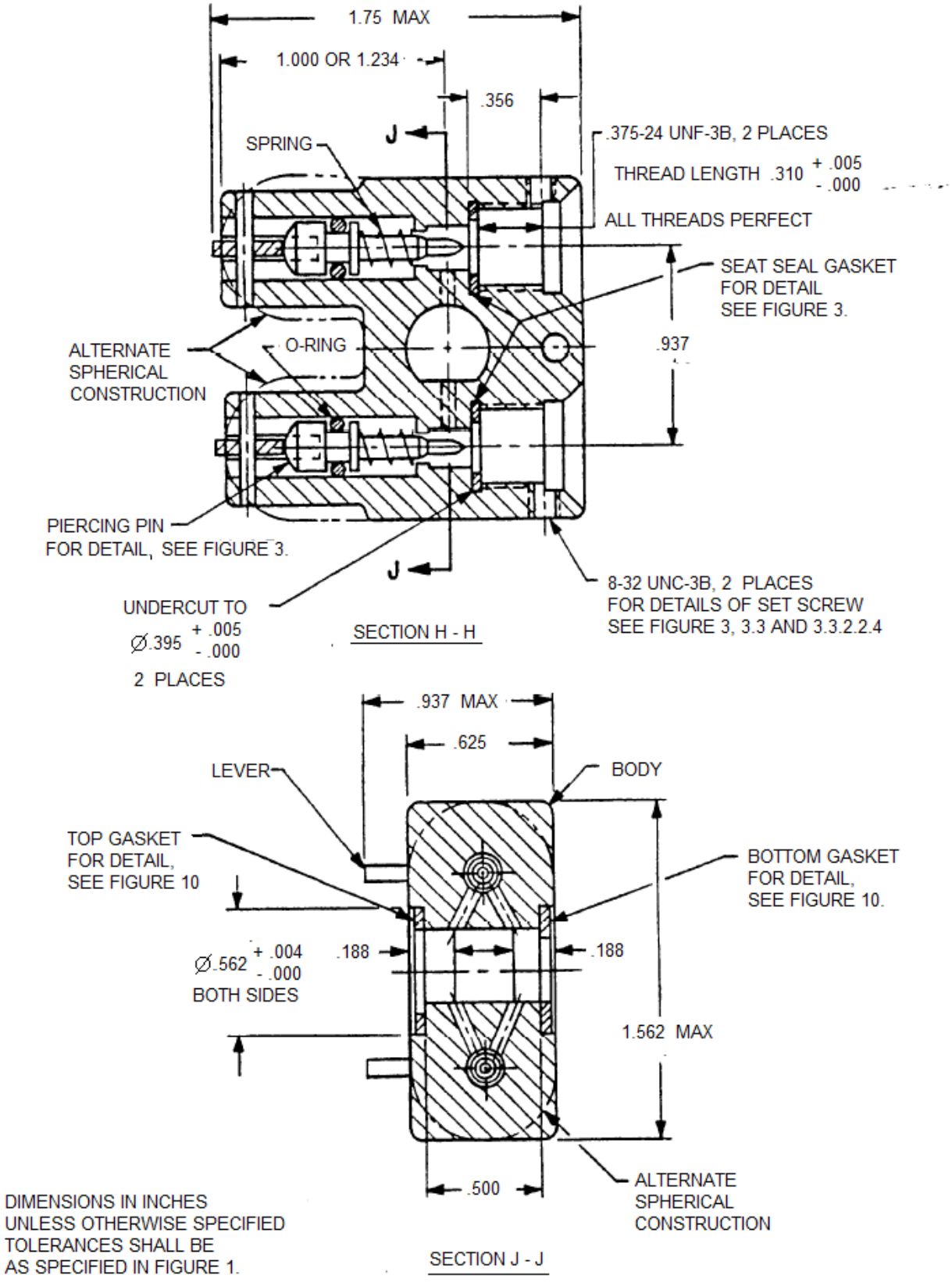
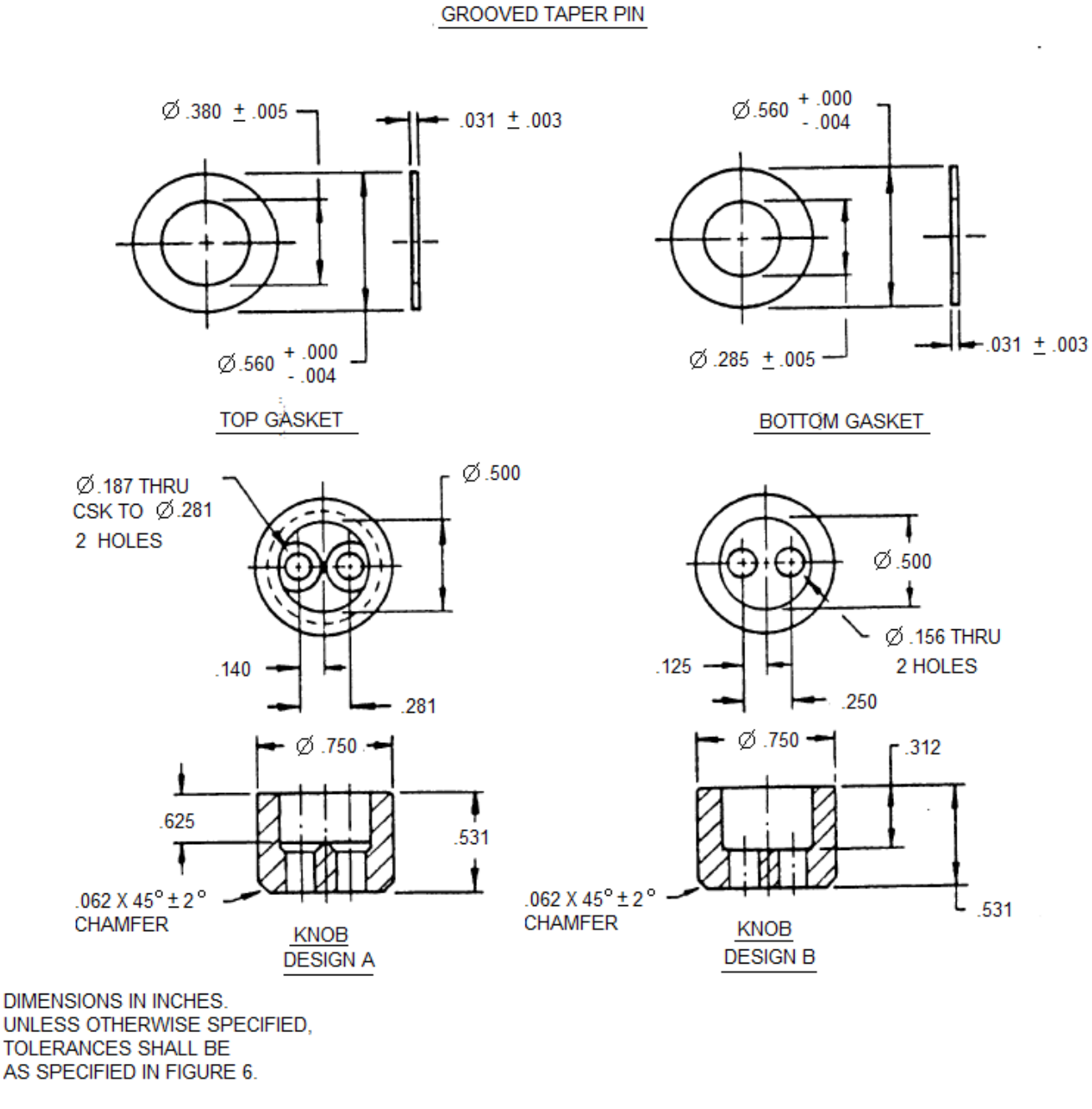


FIGURE 9. SECTIONS H - H AND J - J FOR TYPE III INFLATION ASSEMBLY

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES SHALL BE
AS SPECIFIED IN FIGURE 6.

FIGURE 10. GROOVED TAPER PIN, TOP GASKET, BUTTON GASKET, AND KNOB DETAILS

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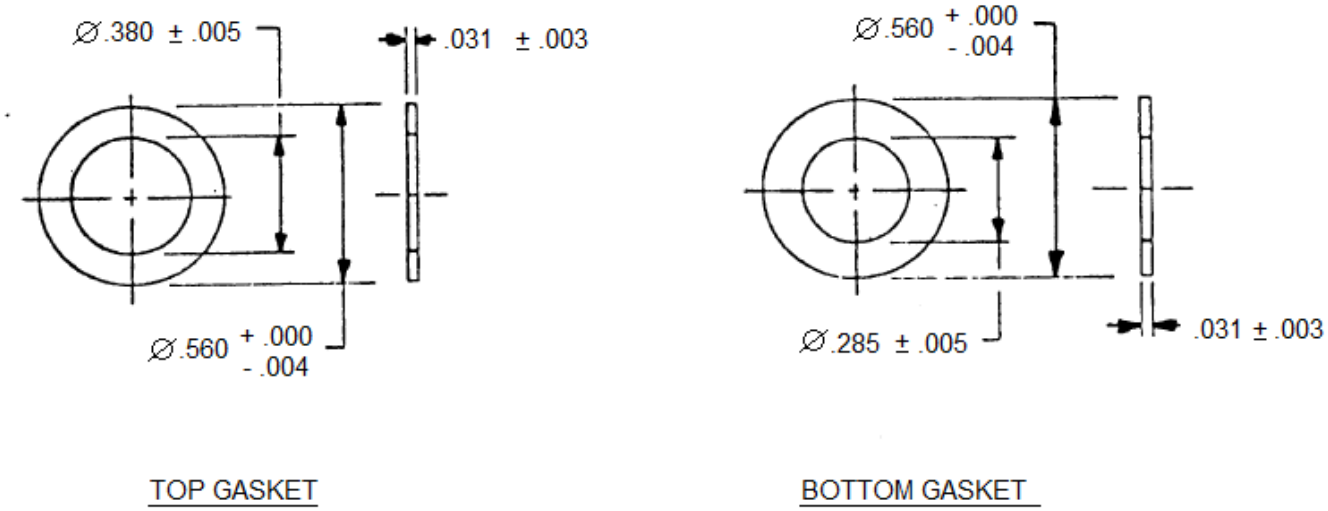
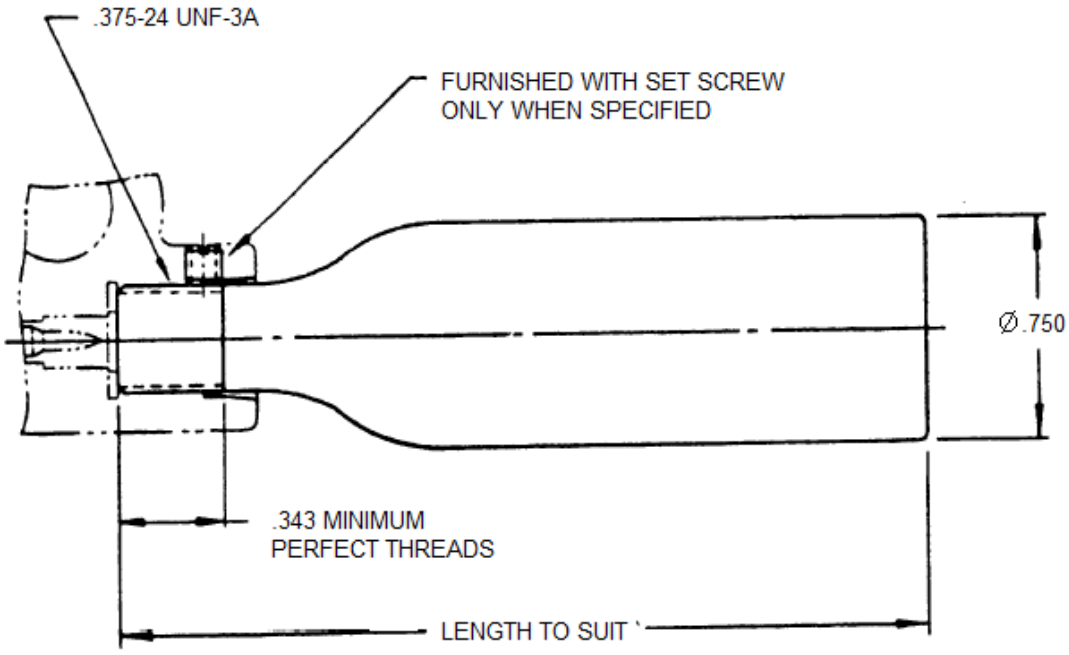
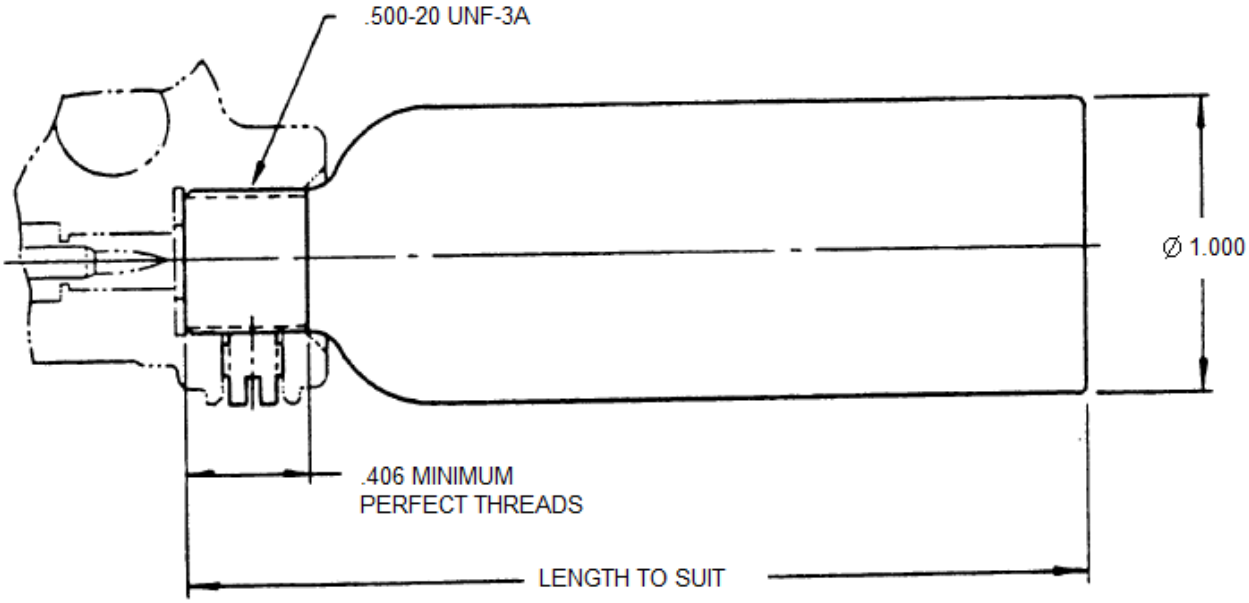


FIGURE 11. MANIFOLD STEM GASKET

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STANDARD GAGE FOR TYPE I AND III INFLATION ASSEMBLIES



STANDARD GAGE FOR TYPE II INFLATION ASSEMBLY

DIMENSIONS IN INCHES.

FIGURE 12. TYPICAL EXAMPLE OF STANDARD CYLINDER GAGE FOR TYPE I, II, AND III INFLATION ASSEMBLIES

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Navy - AS
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Preparing activity:
DLA - IS

(Project 4220-2018-001)

Reviewer:
Army - AV
Navy - MC, SH, YD
Air Force - 11, 99

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